No need to submit. Please attempt all these questions yourself and reach out for any difficulties. Most of these questions are from from [2] and [1] (available here)

1. If A, B, C are finite sets and $|A|$ denotes the number of elements in A you might have seen that:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

   (a) Draw the Venn Diagram for the statement above.

   (b) How does this statement generalize when you have four sets A, B, C, D? Is it possible to draw a Venn Diagram for this?

2. In each of the following statements, if we replace S by one of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, then we get a true statement. Find the appropriate set in each case:

   1. $\{x \in S \mid x^3 = 5\} = \emptyset$
   2. $\{x \in S \mid -1 \leq x \leq 1\} = \{1\}$
   3. $\{x \in S \mid 2 < x^2 < 5\} \setminus \{x \in S \mid x > 0\} = \{-2\}$
   4. $\{x \in S \mid 1 < x \leq 4\} = \{x \in S \mid x^2 = 4\} \cup \{3, 4\}$
   5. $\{x \in S \mid 4x^2 = 1\} \setminus \{x \in S \mid x < 0\} = \{x \in S \mid 5x^2 = 3\} \cup \{x \in S \mid 2x = 1\} \neq \emptyset$.

3. The equation $x + y = z$ has many solutions where $x, y, z \in \mathbb{N}$; the equation $x^2 + y^2 = z^2$ has solutions including $x = 3, y = 4, z = 5$. Let $F = \{n \in \mathbb{N} \mid x^n + y^n = z^n$ has a solution where $x, y, z \in \mathbb{N}\}$. What must be done to show $F = \{1, 2\}$? *Of course you do not have to prove/show this !*

4. Let S be the set of circles in the plane, and let $f : S \mapsto \mathbb{R}$ be defined by $f(S) =$ the area of S. Is f injective? Surjective? Bijective?

   Now let T be the set of circles in the plane whose centre is the origin, and let $\mathbb{R}^+ = \{x \in \mathbb{R} \mid x \geq 0\}$, and define $g : T \mapsto \mathbb{R}^+$ by

$$g(T) = \text{the length of the circumference of } T.$$

   Is g injective? Surjective? Bijective?

5. If A has n elements and B has m elements $n, m \in N$, find the number of functions from A to B. How many of these functions are one-one ? How many are onto? How many are bijective ?

6. If $f : A \to B$, show that for $X \subseteq A$ and $Y \subseteq B$, the formulas $\hat{f}(X) = \{f(x) \mid x \in X\}$, $\tilde{f}(Y) = \{x \in A \mid f(x) \in Y\}$. define two functions $\hat{f} : \mathcal{P}(A) \to \mathcal{P}(B)$, $\tilde{f} : \mathcal{P}(B) \to \mathcal{P}(A)$, where $\mathcal{P}(X)$ denotes the set of all subsets of X.
   Prove that for all $X_1, X_2 \subseteq A$ and $Y_1, Y_2 \subseteq B$:

   1. $\hat{f}(X_1 \cup X_2) = \hat{f}(X_1) \cup \hat{f}(X_2)$
   2. $\hat{f}(X_1 \cap X_2) \subseteq \hat{f}(X_1) \cap \hat{f}(X_2)$, but equality need not hold
   3. $\tilde{f}(Y_1 \cup Y_2) = \tilde{f}(Y_1) \cup \tilde{f}(Y_2)$
   4. $\tilde{f}(Y_1 \cap Y_2) = \tilde{f}(Y_1) \cap \tilde{f}(Y_2)$

   Can we improve (b) to equality if f is known to be surjective? Injective? Bijective?

   The following functions are to be defined so that their codomain is $\mathbb{R}$, and their domains are certain subsets of $\mathbb{R}$. In each case, determine the largest possible domain.

   (a) $f(x) = \log x$

   (b) $f(x) = \log \log \cos x$

(c) $f(x) = -x$

(d) $f(x) = \log(1 - x^2)$

(e) $f(x) = \log(\sin^2(x))$

(f) $f(x) = e^{x^2}$

(g) $f(x) = \frac{1}{x^2-1}$

(h) $f(x) = \sqrt{(x-1)(x-2)(x-3)(x-4)(x-5)(x-6)}$ (positive square root).

7. Problems 2.1-2.6 from [1]

8. (*Optional*) The following are axioms for a (hitherto undefined) mathematical structure known as a bureaucracy. This consists of:

   - a set B of bureaucrats,
   - a set C of committees,
   - a relation S between B and C (read serves on),

   satisfying the following axioms:

   (B1) Every bureaucrat serves on at least three different committees.
   (B2) Every committee is served on by at least three different bureaucrats.
   (B3) Given two distinct committees, exactly one bureaucrat serves on both.
   (B4) Given two distinct bureaucrats, there is exactly one committee on which they both serve.

   Prove from these axioms that if the number of bureaucrats is finite, so is the number of committees. Prove that there are always at least seven bureaucrats in a bureaucracy, and find a bureaucracy with exactly seven bureaucrats.

References

[1] Stephen Boyd and Lieven Vandenberghe. *Introduction to applied linear algebra: vectors, matrices, and least squares*. 2018.

[2] Ian Stewart and David Tall. *The foundations of mathematics*. OUP Oxford, 2015.